

ICT1011: Computer Organisation and Architecture

Tutorial 1: Computer Architecture Basics

Q1. Reduced Instruction Set Computing

Consider the following instructions and decide whether they are from a RISC or CISC processor:

- a) **MPX:** Multiply the content of two memory locations, then add the result to an accumulator.
- b) **BCDD:** Perform a binary-coded decimal division on two registers, format the result in scientific notation and store as ASCII to a memory block ready for display to the screen.
- c) **ADD:** ADD one operand to another and return the result as a third operand. The operands and result are register contents only.
- d) **LMUL Rc, Ra, Rb:** Perform a 100 clock cycle long multiplication of $Ra * Rb$ and place the result in register Rc.

Q2. Flynn's classification

Michael Flynn was the first to classify machines based upon how they handle multiple items of data in relation to their instruction words. Consider how he would rate the following (mainly fictitious) types of instruction that different processors and systems might implement:

Q2a) ADD R1, R2, R3 ; perform addition $R1 = R2 + R3$

Q2b) MADD {R0-R7}, {R8-R15}, #0x10 ; $R0=R8+16, R1=R8+16, R2=R9+16... R7=R15+16$

Q2c) ADD R0, R1, R2 || SUB R4, R5, R6 ; performs $R0=R1+R2, R4=R5-R6$ simultaneously

Q3. Memory Bus Architecture

Fig. Q3 shows three CPUs each of which connect to memory in slightly different ways. Classify CPU A, B and C into von Neumann or Harvard architectures and rank the three in terms of:

- (i) system cost,
- (ii) device cost,
- (iii) ease of programming,
- (iv) CPU pin count requirements,
- (v) performance.

Extension: modern PCs are highly graphical. Can you guess which memory connection method is used on most modern PCs to prevent display updates from disrupting normal program memory access?

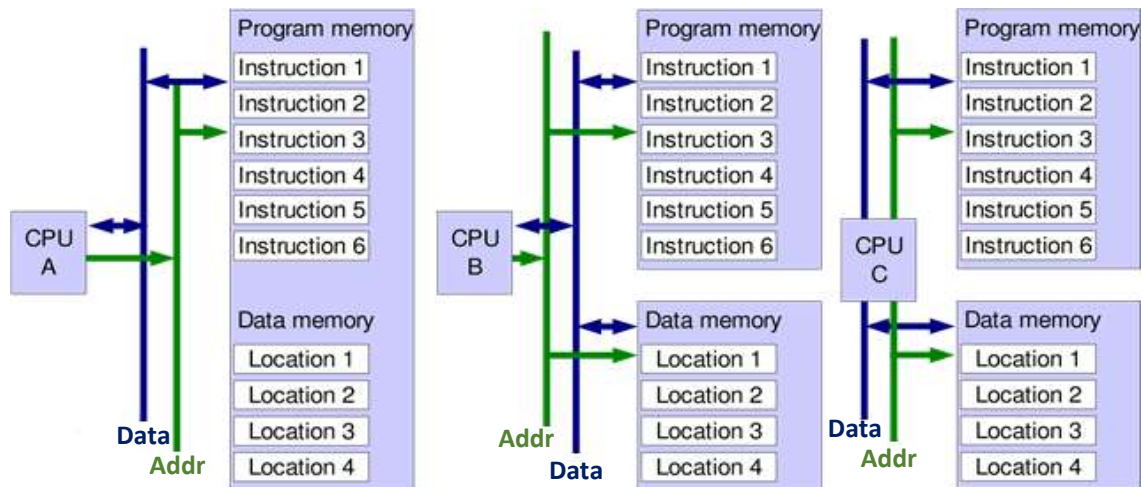


Fig. Q3

Q4. CPU internal bus arrangements

Consider the three types of internal CPU bus arrangements below, all from RISC CPUs. The buses shown in the diagram are used to transfer data. Assume that all the buses are integrated with the appropriate control signals required to enable the data transfers. One clock cycle is required to transfer the contents of one register across the bus. The ALU accepts two input operands and generates one output result. All inputs and output of the ALU are buffered by intermediate registers.

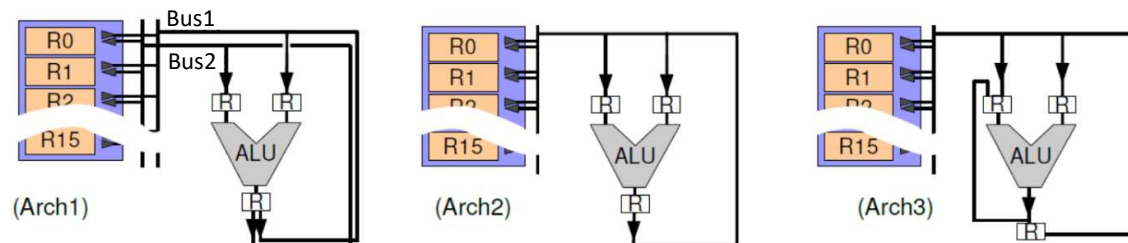


Fig. Q4

Answer the following below, giving a brief explanation to support the reasoning behind your answer:

- Q4a) Which of these architectures (if any) can simultaneously transfer two different operands from registers into the ALU during the same clock cycle?
- Q4b) Which of these architectures (if any) can execute $R0 = R1 + R2$ in a single clock cycle?
- Q4c) Which of these architectures will be most efficient to perform the following operation:

$$y = \sum_{k=0}^{10} x(k)$$

Assume that the x values are already stored in registers, so that R0 contains $x(0)$, R1 contains $x(1)$... R10 contains $x(10)$.